

## T3 group factors: UV

Active theory: SM +  $S_1 + S_2 + F$ . The same beta function is shown across runs in a single comparison table, with exact analytic group factors in the run columns.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table,  $\mathcal{S}$  denotes the RGE structure multiplying the displayed group factor in  $16\pi^2\beta$ .

## Notation key

| symbol                    | interaction / definition                                           |
|---------------------------|--------------------------------------------------------------------|
| $g_1$                     | $D_\mu \supset ig_1 Y B_\mu$                                       |
| $g_2$                     | $D_\mu \supset ig_2 T^A W_\mu^A$                                   |
| $g_3$                     | $D_\mu \supset ig_3 t^A G_\mu^A$                                   |
| $Y_u$                     | $-\bar{Q}Y_u\tilde{H}u + \text{h.c.}$                              |
| $Y_d$                     | $-\bar{Q}Y_dHd + \text{h.c.}$                                      |
| $Y_e$                     | $-\bar{L}Y_eHe + \text{h.c.}$                                      |
| $y_1$                     | $LF S_1 + \text{h.c.}$                                             |
| $y_2$                     | $LF S_2 + \text{h.c.}$                                             |
| $M_F$                     | $-\bar{F}M_F F$                                                    |
| $T$                       | $\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$ |
| $T_\nu^{(1)}$             | $\text{Tr}(y_1^\dagger y_1)$                                       |
| $T_\nu^{(2)}$             | $\text{Tr}(y_2^\dagger y_2)$                                       |
| $T_\nu$                   | $T_\nu^{(1)} + T_\nu^{(2)}$                                        |
| $m_1^2$                   | $m_1^2 S_1^\dagger S_1$                                            |
| $m_2^2$                   | $m_2^2 S_2^\dagger S_2$                                            |
| $\lambda_1$               | $\frac{1}{2}\lambda_1 (H^\dagger H)^2$                             |
| $\lambda_{S_1}^{(1)}$     | $\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$               |
| $\lambda_{S_2}^{(1)}$     | $\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$               |
| $\lambda_{HS_1}^{(1)}$    | $\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$              |
| $\lambda_{HS_2}^{(1)}$    | $\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$              |
| $\lambda_{HS_1}^{(A)}$    | $\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$      |
| $\lambda_{HS_2}^{(A)}$    | $\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$      |
| $\lambda_5$               | $\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$                      |
| $\lambda_{12}^{(1)}$      | $\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$            |
| $\lambda_{12}^{(A)}$      | $\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$    |
| $\lambda_{12}^{(\times)}$ | $\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$       |
| $\lambda_{S_1}^{(A)}$     | $\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$   |
| $\lambda_{S_2}^{(A)}$     | $\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$   |
| $\alpha$                  | $Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$   |
| $C_2(d)$                  | $(d^2-1)/4$                                                        |
| $T(d)$                    | $d(d^2-1)/12$                                                      |
| $C_{LLS_1S_2}^{ijAB}$     | $L_i L_j S_1^A S_2^B$                                              |
| $C_{ijab}$                | $L_i L_j \phi_a \phi_b$                                            |
| $C_5^{ij}$                | $(L_i^T C L_j) H H$                                                |
| $R_W$                     | $16\pi^2\beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$            |

## Field representations

| run    | $S_1$               | $S_2$               | $F$       | $\alpha$ |
|--------|---------------------|---------------------|-----------|----------|
| T3 - C | $(2, -\frac{3}{2})$ | $(2, -\frac{1}{2})$ | $(3, -1)$ | -3       |

# Gauge and Yukawa beta functions

$\beta_{g_1}^{(1)}$

| $\mathcal{S}$ | T3 - C ( $\alpha = -3$ ) |
|---------------|--------------------------|
| $g_1^3$       | $\frac{41}{2}$           |

$\beta_{g_2}^{(1)}$

| $\mathcal{S}$ | T3 - C ( $\alpha = -3$ ) |
|---------------|--------------------------|
| $g_2^3$       | $\frac{31}{6}$           |

$\beta_{g_3}^{(1)}$

| $\mathcal{S}$ | T3 - C ( $\alpha = -3$ ) |
|---------------|--------------------------|
| $g_3^3$       | $-7$                     |

$\beta_{Y_u}^{(1)}$

| $\mathcal{S}$                 | T3 - C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $(Y_d(Y_d)^\dagger Y_u)_{ij}$ | $-\frac{3}{2}$           |
| $(Y_u(Y_u)^\dagger Y_u)_{ij}$ | $\frac{3}{2}$            |
| $g_2^2(Y_u)_{ij}$             | $-\frac{9}{4}$           |
| $g_3^2(Y_u)_{ij}$             | $-8$                     |
| $g_1^2(Y_u)_{ij}$             | $-\frac{17}{12}$         |
| $T(Y_u)_{ij}$                 | $1$                      |

$\beta_{Y_d}^{(1)}$

| $\mathcal{S}$                 | T3 - C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $(Y_d(Y_d)^\dagger Y_d)_{ij}$ | $\frac{3}{2}$            |
| $(Y_u(Y_u)^\dagger Y_d)_{ij}$ | $-\frac{3}{2}$           |
| $g_2^2(Y_d)_{ij}$             | $-\frac{9}{4}$           |
| $g_3^2(Y_d)_{ij}$             | $-8$                     |
| $g_1^2(Y_d)_{ij}$             | $-\frac{5}{12}$          |
| $T(Y_d)_{ij}$                 | $1$                      |

$\beta_{Y_e}^{(1)}$

| $\mathcal{S}$                 | T3 - C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $(y_1(y_1)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            |
| $(y_2(y_2)^\dagger Y_e)_{ij}$ | $\frac{3}{4}$            |
| $(Y_e(Y_e)^\dagger Y_e)_{ij}$ | $\frac{3}{2}$            |
| $g_2^2(Y_e)_{ij}$             | $-\frac{9}{4}$           |
| $g_1^2(Y_e)_{ij}$             | $-\frac{15}{4}$          |
| $T(Y_e)_{ij}$                 | $1$                      |

$\beta_{y_1}^{(1)}$ 

| $\mathcal{S}$                 | T3 – C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $T_\nu^{(1)}(y_1)_{ij}$       | $\frac{3}{2}$            |
| $(y_2(y_2)^\dagger y_1)_{ij}$ | $\frac{3}{4}$            |
| $(Y_e(Y_e)^\dagger y_1)_{ij}$ | $\frac{1}{2}$            |
| $(y_1(y_1)^\dagger y_1)_{ij}$ | $\frac{5}{4}$            |
| $g_2^2(y_1)_{ij}$             | $-\frac{33}{4}$          |
| $g_1^2(y_1)_{ij}$             | $-\frac{15}{4}$          |

 $\beta_{y_2}^{(1)}$ 

| $\mathcal{S}$                 | T3 – C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $T_\nu^{(2)}(y_2)_{ij}$       | $\frac{3}{2}$            |
| $(y_1(y_1)^\dagger y_2)_{ij}$ | $\frac{3}{4}$            |
| $(Y_e(Y_e)^\dagger y_2)_{ij}$ | $\frac{1}{2}$            |
| $(y_2(y_2)^\dagger y_2)_{ij}$ | $\frac{5}{4}$            |
| $g_2^2(y_2)_{ij}$             | $-\frac{33}{4}$          |
| $g_1^2(y_2)_{ij}$             | $-\frac{15}{4}$          |

 $\beta_{M_F}$ 

| $\mathcal{S}$          | T3 – C ( $\alpha = -3$ ) |
|------------------------|--------------------------|
| $(y_1^T y_1^*) M_F$    | $\frac{1}{2}$            |
| $M_F(y_2^\dagger y_2)$ | $\frac{1}{2}$            |
| $g_2^2 M_F$            | $-12$                    |
| $g_1^2 M_F$            | $-6$                     |

 $\beta_{m_1^2}$ 

| $\mathcal{S}$                 | T3 – C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $m_1^2 T_\nu^{(1)}$           | $3$                      |
| $M_F^\dagger M_F T_\nu^{(1)}$ | $-6$                     |
| $\lambda_{S_1}^{(1)} m_1^2$   | $6$                      |
| $g_2^2 m_1^2$                 | $-\frac{9}{2}$           |
| $g_1^2 m_1^2$                 | $-\frac{27}{2}$          |
| $\lambda_{12}^{(1)} m_2^2$    | $4$                      |

 $\beta_{m_2^2}$ 

| $\mathcal{S}$                 | T3 – C ( $\alpha = -3$ ) |
|-------------------------------|--------------------------|
| $m_2^2 T_\nu^{(2)}$           | $3$                      |
| $M_F^\dagger M_F T_\nu^{(2)}$ | $-6$                     |
| $\lambda_{S_2}^{(1)} m_2^2$   | $6$                      |
| $g_2^2 m_2^2$                 | $-\frac{9}{2}$           |
| $g_1^2 m_2^2$                 | $-\frac{3}{2}$           |
| $\lambda_{12}^{(1)} m_1^2$    | $4$                      |

$\beta_{\lambda_{HS_1}^{(1)}}$ 

| $\mathcal{S}$                                | T3 - C ( $\alpha = -3$ ) |
|----------------------------------------------|--------------------------|
| $\lambda_{HS_1}^{(1)} T_\nu^{(1)}$           | 3                        |
| $\text{Tr}(y_1^\dagger y_1 Y_e^\dagger Y_e)$ | -3                       |
| $ \lambda_5 ^2$                              | 6                        |
| $g_2^2 \lambda_{HS_1}^{(1)}$                 | -9                       |
| $g_1^2 \lambda_{HS_1}^{(1)}$                 | -15                      |
| $\lambda_1 \lambda_{HS_1}^{(1)} 2$           | 6                        |
| $\left(\lambda_{HS_1}^{(1)}\right)^2$        | 4                        |
| $\lambda_{HS_1}^{(1)} \lambda_{S_1}^{(1)}$   | 6                        |
| $\lambda_{12}^{(1)} \lambda_{HS_2}^{(1)}$    | 4                        |
| $g_2^4$                                      | $\frac{9}{4}$            |
| $g_1^4$                                      | $\frac{27}{4}$           |
| $\left(\lambda_{HS_1}^{(A)}\right)^2$        | $\frac{3}{4}$            |
| $\lambda_{HS_1}^{(1)} T$                     | 2                        |

 $\beta_{\lambda_{HS_2}^{(1)}}$ 

| $\mathcal{S}$                                | T3 - C ( $\alpha = -3$ ) |
|----------------------------------------------|--------------------------|
| $\lambda_{HS_2}^{(1)} T_\nu^{(2)}$           | 3                        |
| $\text{Tr}(y_2^\dagger y_2 Y_e^\dagger Y_e)$ | -3                       |
| $ \lambda_5 ^2$                              | 6                        |
| $g_2^2 \lambda_{HS_2}^{(1)}$                 | -9                       |
| $g_1^2 \lambda_{HS_2}^{(1)}$                 | -3                       |
| $\lambda_1 \lambda_{HS_2}^{(1)} 2$           | 6                        |
| $\left(\lambda_{HS_2}^{(1)}\right)^2$        | 4                        |
| $\lambda_{HS_2}^{(1)} \lambda_{S_2}^{(1)}$   | 6                        |
| $\lambda_{12}^{(1)} \lambda_{HS_1}^{(1)}$    | 4                        |
| $g_2^4$                                      | $\frac{9}{4}$            |
| $g_1^4$                                      | $\frac{3}{4}$            |
| $\left(\lambda_{HS_2}^{(A)}\right)^2$        | $\frac{3}{4}$            |
| $\lambda_{HS_2}^{(1)} T$                     | 2                        |

 $\beta_{\lambda_{12}^{(1)}}$ 

| $\mathcal{S}$                                | T3 - C ( $\alpha = -3$ ) |
|----------------------------------------------|--------------------------|
| $\text{Tr}(y_1^\dagger y_1 y_2^\dagger y_2)$ | $-\frac{9}{2}$           |
| $ \lambda_5 ^2$                              | 3                        |
| $g_2^2 \lambda_{12}^{(1)}$                   | -9                       |
| $g_1^2 \lambda_{12}^{(1)}$                   | -15                      |
| $\left(\lambda_{12}^{(1)}\right)^2$          | 4                        |
| $\lambda_{12}^{(1)} \lambda_{S_1}^{(1)}$     | 6                        |
| $\lambda_{12}^{(1)} \lambda_{S_2}^{(1)}$     | 6                        |
| $\lambda_{HS_1}^{(1)} \lambda_{HS_2}^{(1)}$  | 4                        |
| $g_2^4$                                      | $\frac{9}{4}$            |
| $g_1^4$                                      | $\frac{27}{4}$           |
| $\left(\lambda_{12}^{(A)}\right)^2$          | $\frac{3}{4}$            |

|                            |   |
|----------------------------|---|
| $\lambda_{12}^{(1)} T_\nu$ | 3 |
|----------------------------|---|

$\beta_{\lambda_5}$

| $\mathcal{S}$                    | T3 - C ( $\alpha = -3$ ) |
|----------------------------------|--------------------------|
| $\lambda_5 \lambda_1$            | 2                        |
| $\lambda_5 \lambda_{HS_1}^{(1)}$ | 4                        |
| $\lambda_5 \lambda_{HS_2}^{(1)}$ | 4                        |
| $\lambda_5 \lambda_{12}^{(1)}$   | 2                        |
| $g_2^2 \lambda_5$                | -9                       |
| $g_1^2 \lambda_5$                | -9                       |
| $\lambda_5 \lambda_{HS_1}^{(A)}$ | -2                       |
| $\lambda_5 \lambda_{HS_2}^{(A)}$ | 2                        |
| $\lambda_5 \lambda_{12}^{(A)}$   | $-\frac{1}{2}$           |
| $\lambda_5 T$                    | 2                        |
| $\lambda_5 T_\nu$                | $\frac{3}{2}$            |

## Additional scalar-coupling beta functions

$\beta_{\lambda_{HS_1}^{(A)}}$

| $\mathcal{S}$ | T3 - C ( $\alpha = -3$ ) |
|---------------|--------------------------|
| $g_1^2 g_2^2$ | -18                      |

$\beta_{\lambda_{HS_2}^{(A)}}$

| $\mathcal{S}$ | T3 - C ( $\alpha = -3$ ) |
|---------------|--------------------------|
| $g_1^2 g_2^2$ | -6                       |

$\beta_{\lambda_{12}^{(A)}}$

| $\mathcal{S}$ | T3 - C ( $\alpha = -3$ ) |
|---------------|--------------------------|
| $g_1^2 g_2^2$ | 18                       |

## First threshold

The first EFT GroupFactors report below is selected from the actual threshold plan stored in the run. No assumption is made here about whether  $F$ ,  $S_1$ ,  $S_2$ , or a degenerate set is integrated out first.